

NOISY ARCHIVE DOCUMENT 02 — HISTORY / TECHNOLOGY / NATURE

INTENTIONALLY NOISY REFERENCE FILE: duplicated ideas, OCR-like defects, irrelevant notes, tables, inconsistent spacing and mixed formatting are included for retrieval testing.

Record	ARC-78414	Quality	PARTIAL / noisy
Collection	General knowledge	Revision	2.7
Language	English	Source type	mixed

Ancient & Modern History

01. Mesopotamia is a historical region associated with the Tigris and Euphrates river systems. Early cities and writing developed there.
02. Ancient Egypt depended heavily on the Nile for agriculture, transportation and settlement.
03. The Indus Valley Civilization included planned settlements and sophisticated drainage systems.
04. Ancient Greek city-states used different political systems. Athens developed a form of direct citizen participation.
05. Rome expanded into a large Mediterranean empire through military, political and administrative systems.
06. The printing press greatly increased the speed and scale of text reproduction in Europe during the fifteenth century.

Unrelated editor note: Previous archive copies used inconsistent terminology. This note may be irrelevant to the question being asked.

Computing

07. A bit is a binary digit. Eight bits commonly make one byte.
08. RAM is generally volatile memory, while SSDs and hard drives provide non-volatile storage.
09. A CPU executes instructions and performs arithmetic, logical and control operations. Modern CPUs may contain multiple cores.
10. HTTP is used by the Web. HTTPS adds TLS protection for confidentiality and authentication.
11. Databases store and retrieve information using structured systems. SQL is widely used for relational databases.
12. Machine learning systems learn patterns from data. Data quality, evaluation and deployment conditions affect results.

Extracted table — some labels may be OCR-shifted

Item	Value	Unit	Note
Water coverage	71	%	approx.
Heart chambers	4	count	standard
Light speed	299792458	m/s	exact SI
Everest	8849	m	approx.
Moon period	29.5	days	synodic

Animals & Plants

13. Blue whales are the largest known animals. Their marine environment supports very large body sizes.
14. Bats are mammals and the only mammals capable of sustained powered flight.
15. Octopuses have three hearts and use copper-containing hemocyanin to transport oxygen.
16. Bees are important pollinators, but wind, birds, bats and many other insects also pollinate plants.
17. Fungi obtain nutrients by external digestion and absorption and are important decomposers.
18. Trees transport water through vascular tissues and perform much of their photosynthesis in leaves.

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Materials & Engineering

19. Steel is an iron-based alloy containing carbon and often other elements. Composition affects strength and other properties.
20. Copper is an excellent electrical conductor and is widely used in wiring and electrical equipment.
21. Concrete commonly contains cement, water and aggregates. Curing strongly affects its final properties.
22. Aluminium is relatively light and develops a protective oxide layer. It is widely used in transport and construction.
23. Glass is generally an amorphous solid. Its composition can be adjusted for optical, thermal and mechanical properties.

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Everyday Science

24. Boiling occurs when a liquid's vapour pressure equals the surrounding pressure, so boiling temperature changes with pressure.
25. Evaporation occurs at a liquid surface and can happen below the boiling point.
26. Ice is less dense than liquid water near ordinary conditions, allowing it to float.
27. Sound is a mechanical wave and requires a medium. It can travel through gases, liquids and solids.
28. Rainbows result from refraction, internal reflection and dispersion of light in water droplets.
29. Earth's seasons are primarily caused by axial tilt, which changes sunlight angle and day length during the orbit.

APPENDIX — DUPLICATED EXCERPTS

Mesopotamia is a historical region associated with the Tigris and Euphrates river systems. Early cities and writing developed there.

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The Indus Valley Civilization included planned settlements and sophisticated drainage systems.

OCR / SEARCH TEST NOTES

- The answer to a query may appear in prose and separately in a table.
- Some numbers are approximate and some are exact; context is required.
- Header and footer text can be mistaken for document content.
- Character substitutions such as rn/m and 1/l are intentional.
- Several passages are semantically unrelated to nearby facts.